

Health and Sleep Data Analysis Report

Dataset: Relationship between Health and Sleep (2024)

Source: Kaggle

Task Performed: Exploratory data analysis and insight generation using SQL.

1. Introduction

This paper presents a comprehensive analysis of the interaction between health variables, such as sleep, mental health, stress, physical activity, and demographic variables like age, sex, occupation, and city. The data has a variety of attributes related to the health and lifestyle of people, thus offering a multi-dimensional analysis through SQL queries.

2. Objectives

- To understand sleep patterns across age, gender, and profession.
- To determine correlations among stress, sleep quality, mental well-being, and exercise.
- To identify demographic segments that are at increased risk for poor mental health and sleep.
- To illustrate advanced SQL methods for data profiling and analysis.

3. Key Findings

3.1 Sleep Patterns by Demographics

- Age groups: The sleep duration differs considerably by age groups. Young adults (18–30) have a greater average sleep duration compared to the elderly.
- Gender: There are slight differences in mean sleep quality and mental health scores between females and males.
- Occupation: Certain occupations exhibit markedly reduced sleep quality and increased stress levels that reflect the impact of occupation on health.

3.2 Sleep and Mental Health Correlations

- Those with poor sleep (<6 hours) have lower mental health scores (<5) and are a potential at-risk subgroup.
- Ranking individuals by combined sleep quality and mental health score helps identify those needing attention.

3.3 Physical Activity Impact

- In most people, greater amounts of physical activity are highly associated with better sleep quality.
- There are exceptions of high activity status with poor sleep, however, indicating complexity in lifestyle-health interactions.

3.4 Geographic Disparities

- Cities differ in mean sleep quality and mental health rating, suggesting potential environmental or cultural factors.

4. SQL Techniques Employed

- Conditional aggregation: Employing CASE statements to categorize age and sleep types.
- Window operations: RANK and PERCENTILE_CONT used to rank the individuals and determine median sleep duration.
- Aggregate operators: AVG, COUNT with GROUP BY and HAVING for filtering and aggregation.
- Views: Employed reusable SQL views for sleep summaries to enhance query modularity.
- Filtering: Advanced WHERE clauses to separate data according to several criteria.

5. Sample SQL Queries Overview

QUESTIONS
Average sleep duration by age group
Active individuals with good sleep percentage
Get View creation for sleep-health summary
Calculation of median sleep duration (MySQL 8+)
Jobs with below-average sleep quality

6. Conclusion

This report emphasizes the potential of SQL to conduct exploratory data analysis for health datasets. Through the analysis of sleep-related variables, mental health variables, and lifestyle variables, it gives solid indications of population health trends. The queries illustrate the actual application of SQL potential to profile, filter, and rank people for generating actionable health advice.